Route-1 Cubic Size Bound:
How the Four Lanes Fit Together

Connecting map — COVER / BND / SEQ / FORMALIZE

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0. Orientation

There are *two independent* results about the strong simplifier $S = \texttt{rsimpStrong\_raw}$: **(i) correctness** (the strong lexer returns the POSIX value) and **(ii) the cubic size bound**. They are proved *separately*; (i) may inspire (ii)'s design but is *not* assumed by it. **This note maps only track (ii)** and shows where each of the four Codex lanes sits inside its decomposition. Every box below names the exact Isabelle lemma so each lane's own summary can be located in the whole.

Primitives (self-contained). All equations are the verbatim Isabelle definitions, in symbols. Regexes r, t, k over Σ on the clean fragment:

$$r \ ::= \ \mathbf{0} \mid \mathbf{1} \mid c\,(c \in \Sigma) \mid r \cdot t \mid \texttt{RALTS}\, rs \ (rs \text{ a list}) \mid r^*,$$

with $\operatorname{rsize} \mathbf{0} = \operatorname{rsize} \mathbf{1} = \operatorname{rsize} c = 1$, $\operatorname{rsize}(\texttt{RALTS}\, rs) = 1 + \sum_i \operatorname{rsize}(rs_i)$, $\operatorname{rsize}(rt) = 1 + \operatorname{rsize} r + \operatorname{rsize} t$, $\operatorname{rsize}(r^*) = 1 + \operatorname{rsize} r$, and set-size $\operatorname{rsize}_s(V) = \sum_{q \in V} \operatorname{rsize} q$. $\texttt{apder\_nf}$ = structural normal form (no $\mathbf{0}/\mathbf{1}$ factor in a sequence, no nested/redundant alternation); $\texttt{apder\_clean} = \texttt{apder\_nf}$ on this fragment — the regime one may assume.

Frontier rf (split a top alternation, else a singleton):

$$\operatorname{rf}(\mathbf{0}) = \varnothing, \qquad \operatorname{rf}(\texttt{RALTS}\, rs) = \textstyle\bigcup_i \operatorname{rf}(rs_i), \qquad \operatorname{rf}(r) = \{r\} \text{ otherwise.}$$

Plugs σ_4 (weak), σ_7 (strong): $\sigma_4(r, k)$ appends continuation k after the sequence-atom r, right-associating and eliding units/zeros,

$$\sigma_4(\mathbf{1}, k) = k, \quad \sigma_4(\mathbf{0}, k) = \mathbf{0}, \quad \sigma_4(r_1 \cdot r_2, k) = \sigma_4(r_1, \sigma_4(r_2, k)), \quad \sigma_4(h, \mathbf{1}) = h, \ \ \sigma_4(h, k) = h \cdot k$$

for h a char/$\texttt{RALTS}$/*-headed atom (and $\sigma_4(h, \mathbf{0}) = \mathbf{0}$). The strong plug adds one collapse:

$$\sigma_7(a^*, a^*) = a^*, \quad \sigma_7(a^*, a^* \cdot k) = a^* \cdot k, \quad \sigma_7 = \sigma_4 \text{ otherwise.}$$

Strong normaliser $S = \texttt{rsimpStrong\_raw}$ (recursive); $S\,\mathbf{0} = \mathbf{0}$, $S\,\mathbf{1} = \mathbf{1}$, $S\,c = c$, and

$$S(r_1 \cdot r_2) = \sigma_7(Sr_1, Sr_2), \quad S(\texttt{RALTS}\, rs) = \operatorname{pr}\big(\operatorname{flat}(\langle S\, rs_i \rangle_i)\big), \quad S(r^*) = \begin{cases} \mathbf{1}, & Sr \in \{\mathbf{0}, \mathbf{1}\} \\ (Sr)^*, & \text{else} \end{cases}$$

where pr is the *set-level cross-row prune*: in a later row $(\texttt{RALTS}\, lrs) \cdot k$ it deletes the head-branches already present in an earlier row with the *same* k, re-plugging by σ_7. pr only *shrinks* rows; its leftover "$s^*{\cdot}s^*$" survivors (uncollapsed, since σ_7 collapses only a *leading* equal-star) are the recurring wall.

Opening $\mathrm{dl} =$ `row_dlforms` (a regex $\to$ its set of linear rows):

$$\mathrm{dl}(\mathbf{0}) = \varnothing, \quad \mathrm{dl}(\texttt{RALTS}\, rs) = \bigcup_i \mathrm{dl}(rs_i), \quad \mathrm{dl}\left((\texttt{RALTS}\, ps){\cdot}k\right) = \bigcup_p \mathrm{dl}(\sigma_7(p,k)), \quad \mathrm{dl}(r) = \mathrm{rf}(r) \text{ else.}$$

Strong opening of a set $C(V) = \bigcup_{p\in V} \mathrm{dl}(S\,p)$ (S *inside* the union $\Rightarrow$ C distributes over V).

Term-frontier accumulator $\mathrm{acc} =$ `apder_term_frontier_acc` (continuation-parametric):

$$\mathrm{acc}(\mathbf{0},k) = \mathrm{acc}(\mathbf{1},k) = \varnothing, \quad \mathrm{acc}(c,k) = \mathrm{rf}(k), \quad \mathrm{acc}(\texttt{RALTS}\, rs, k) = \bigcup_i \mathrm{acc}(rs_i, k),$$

$$\mathrm{acc}(r_1{\cdot}r_2, k) = \mathrm{acc}(r_1, \sigma_4(r_2,k)) \cup \mathrm{acc}(r_2,k), \qquad \mathrm{acc}(r^*,k) = \mathrm{acc}\left(r, \sigma_4(r^*,k)\right).$$

Antimirov rows $\texttt{apder\_rows}\, r = \{r\} \cup \mathrm{rf}(r) \cup \bigcup_{q\in\, \texttt{apder\_terms}\, r} \mathrm{rf}(q)$ (the deduped one-step row index set), and the **target universe** $U(r) = C(\texttt{apder\_rows}\, r)$.

Carrier, base, singletons, excess (the single letters used below).

$$A(r,k) = C\big(\mathrm{rf}(\sigma_4(r,k)) \cup \mathrm{acc}(r,k)\big) \quad (= \texttt{strong\_apder\_acc}), \qquad B(k) = A(\mathbf{1},k),$$

$$\mathrm{sr}(q,k) = C\big(\mathrm{rf}(\sigma_4(\texttt{RALTS}[q],k))\big), \quad \mathrm{st}(q,k) = C(\mathrm{acc}(q,k)), \quad A(\texttt{RALTS}[q],k) = \mathrm{sr}(q,k) \cup \mathrm{st}(q,k),$$

$$D(r,k) = \mathrm{card}\left(A(r,k) \setminus B(k)\right), \qquad D_1(q,k) = D(\texttt{RALTS}[q],k).$$

The final-theorem wrappers `afactored1`, `rpder_strong_rows_raw`, `row_dlformss` are *opaque*: they occur only inside the green gate black-box (§3) and are used nowhere else here.

1. The final theorem (the top)

$$\mathrm{rsize}\big(\texttt{row\_dlformss}\ (\texttt{rpder\_strong\_rows\_raw}\ c\ (\texttt{afactored1}\ r\ s))\big) \ \le\ 2\,(\mathrm{rsize}\, r + 3)^3.$$

Plain words: open every one-step strong-pruned derivative row, dedupe, and the total size is cubic in $\mathrm{rsize}\, r$.

2. The gate: reduce "cubic" to one *linear* count

`DirectUniverseCubic.thy` is *green* except for one obligation. With the universe $U(r) = \texttt{apder\_strong\_dlfrontier}\, r =$ $C(\texttt{apder\_rows}\, r)$, the proved gate lemmas (`actual_gate_from_direct_universe_rowlevel`, `universe_le_cubic_rowlevel` `per_row_size_le_quadratic`) give the final theorem *provided*

$$\boxed{\texttt{card\_apder\_strong\_dlfrontier\_le}: \quad \mathrm{card}\, U(r) \ \le\ \mathrm{rsize}\, r + 1} \tag{G}$$

(each member has *quadratic* size; cubic = quadratic size $\times$ *linear* count). Everything below is the proof of the single linear-count obligation (G). **Any** linear bound $C{\cdot}\mathrm{rsize}\, r + D$ also closes it.

3. Bridge: from U to the excess counter D

Three green facts collapse (G) to a statement about D at continuation $k = \mathbf{1}$:

$U(r) \subseteq A(r, \mathbf{1})$ (`..._subset_strong_apder_acc_RONE`), $A(\mathbf{1}, \mathbf{1}) = \{\mathbf{1}\}$ (`strong_apder_acc_RONE_RONE`),

$$\Rightarrow \quad \text{card}\, U(r) \;\le\; \text{card}\, A(r, \mathbf{1}) \;\le\; 1 + D(r, \mathbf{1}) \quad (\texttt{card\_le\_Suc\_card\_Diff\_singleton}).$$

So (G) follows from the **global excess bound**

$$\boxed{\texttt{card\_strong\_apder\_acc\_diff\_base\_le\_rsize} : \quad D(r, k) \;\le\; \text{rsize}\, r \quad (\text{all } k)} \qquad (\Delta)$$

4. The decomposition (how (Δ) splits down to the four lanes)

(Δ) is proved by induction on r (arbitrary k):

- `RZERO`/`RONE`/`RCHAR`: base cases ($D \le$ rsize directly).
- `RSEQ` $r_1 r_2$: the **RSEQ recurrence** (below).
- `RALTS` rs: the **RALTS budget** $D(\texttt{RALTS}\, rs, k) \;\le\; \sum_{q \in rs} \text{rsize}\, q$ (`..._size_budget_spine`), which needs **L1** + **L2**.
- `RSTAR` r: two star helpers (root-row ≤ 1; the $\sigma_4(\texttt{RSTAR})$ frontier identity).

L2 (singleton size). $\boxed{D_1(q, k) \le \text{rsize}\, q}$ (`card_strong_apder_acc_singleton_RALTS_diff_base_le_rsize`), by induction on q. Its `RSEQ` case is exactly the

$$\textbf{RSEQ recurrence:} \qquad D_1(\texttt{RSEQ}\, r_1 r_2,\ k) \;\le\; \text{rsize}\, r_1 + D_1(r_2, k) \quad (\ldots\texttt{\_RSEQ\_rec\_spine}),$$

and *that* recurrence is what consumes the two boundary lemmas **S1** and **seq_head**. Its `RALTS` case re-uses **L1**.

The three leaf obligations the lanes own:

$$\textbf{L1 (cover):} \quad A(\texttt{RALTS}\, rs, k) \;\subseteq\; \bigcup_{q \in rs} A(\texttt{RALTS}[q], k).$$

$$\textbf{S1 (boundary absorb):} \quad \text{card}\big((A(\mathbf{1}, \sigma_4(t, k)) \setminus B(k)) \cup (\text{st}(t, k) \setminus B(k))\big) \le D_1(t, k).$$

$$\textbf{seq\_head:} \quad \text{card}\big((\text{sr}(\texttt{RSEQ}\, h\, t, k) \cup \text{st}(h, \sigma_4(t, k))) \setminus (B(k) \cup B(\sigma_4(t, k)))\big) \le \text{rsize}\, h.$$

Top-to-leaf chain, one line:

$$\text{cubic} \xLeftarrow{\text{gate}} (\text{G}) \xLeftarrow{\text{bridge}} (\Delta) \xLeftarrow{\text{ind. } r} \{\text{RSEQ rec, RALTS budget, RSTAR}\} \Leftarrow \{\mathbf{L1},\ \mathbf{L2}\} \xLeftarrow{\text{ind. } q} \{\mathbf{S1},\ \mathbf{seq\_h}$$

5. The four lanes, placed in the map

Lane	Proves (Isabelle)	Role / method
COVER	`strong_apder_acc_RALTS_singleton_cover` (**L1**)	Feeds the *RALTS budget*. Method: fix-(a), two carriers; the cross-prune-collapsed s^* escapes are recovered by the *acc* carrier, not a branch root.
BND	`boundary_term_absorb` via **S1**	Feeds the *RSEQ recurrence*. Method: the plain subset is false for `RSEQ`-root; use the asymmetric split (RSEQ→S-image + `card_image_le`; non-RSEQ→plain + `card_mono`).
SEQ	`seq_head_core_le_rsize` (**seq_head**)	Feeds the *RSEQ recurrence*. Method: charge rsize h; `single_root_RSEQ_RCHAR_card_le_one` helper; induction on h (uses **L1** on the RALTS-head case).
FORMALIZE	the *spine*: bridge, (Δ) induction, RSEQ-rec assembly, RALTS budget, RSTAR helpers, $\Rightarrow$ (G)	Assembles §3–§4 on `assumes` for L1/S1/seq_head, swapping each in as its lane lands it.

6. Dependencies / merge order

`COVER`(L1) and `BND`(S1) are independent leaves $\Rightarrow$ land in parallel first. `SEQ` depends on *both* (its RSEQ step uses S1; its RALTS-head case uses L1). `FORMALIZE` integrates all three; once L1, S1, seq_head are green it discharges every `assumes`, proves (Δ) $\Rightarrow$(G)$\Rightarrow$ the cubic theorem. Status (track ii): the spine and many helpers are *green*; L1 / S1 / seq_head are *validated with proof skeletons*, in formalization.